

Across the world, we have data centers that are equipped with extensive infrastructure built using servers. Through these data centers, cloud providers offer a wide range of services, including EBS (Elastic Block Store) for scalable storage solutions.

Understanding EBS and Storage Management:

If you need additional storage on your server, you typically have to stop the service, increase the storage, and then restart the service. However, this approach is not efficient. Instead, you can use EBS, which allows you to add extra storage without needing to stop your server.

EBS volumes act as additional storage that you can easily attach to your servers. Here's how you manage EBS volumes:

Steps to Create and Attach an EBS Volume:

1. **List Current Volumes:**
 - Use the `lsblk` command to see all block devices, including your existing volumes.
 - Use the `df -Th` command to view the complete storage usage on your system.
2. **Create a New EBS Volume:**
 - Go to the **Volumes** section in the AWS Management Console.
 - Specify the volume type and size (ranging from 1GB to 16,384GB) based on your storage requirements.
 - Ensure that the volume is created in the same region as your server.
 - Optionally, add tags for easier identification.
 - Click **Create Volume**.
3. **Attach the Volume to Your Server:**
 - Once the volume is created, navigate to the **Volumes** section.
 - Right-click on the volume or select **Actions**, then choose **Attach Volume**.
 - Select the server to which you want to attach the volume.
 - After attachment, use the `df -h` command to verify that the mount is recognized.
 - Check the volume details using `lsblk`.
4. **Mount the Volume:**
 - The newly attached volume needs to be mounted to be usable.

Create a filesystem on the volume using the `mkfs` command:

bash

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```
sudo mkfs.ext4 /dev/xvdf
```

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Create a directory to mount the volume:

bash

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```
sudo mkdir /jithendar
```

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Mount the volume to the directory:

bash

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sudo mount /dev/xvdf /jithendar
```

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Now, you can change to the directory and start using the volume:

bash

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```
cd /jithendar
```

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Sharing Data Between Servers:

EBS volumes are designed to be attached to a single server at a time. If you need to access the same data on a different server, you can create a snapshot of the volume and then create a new volume from that snapshot:

1. Create a Snapshot:

- Go to the **Volumes** section, right-click on the volume, and choose **Create Snapshot**.
- Add a description and tags (optional), and click **Create Snapshot**.
- Check the **Snapshots** section to confirm the snapshot is ready.

2. Create a Volume from the Snapshot:

- In the **Snapshots** section, right-click on the snapshot and choose **Create Volume**.
- Specify the size (which can be larger than the original volume) and create the volume.
- Attach the newly created volume to the second server using the same process as before.

3. Mount the Volume on the Second Server:

Without creating a new filesystem (to avoid overwriting data), create a directory:

bash

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```
sudo mkdir /ramagiri
```

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Mount the volume:

bash

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```
sudo mount -t ext4 /dev/xvdf /ramagiri
```

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- The data from the first server will now be accessible on the second server.

Cross-Region Snapshot Management:

If you need to transfer data across different regions, you can copy snapshots between regions:

1. Copy a Snapshot to Another Region:

- Go to the **Snapshots** section, click **Actions**, and choose **Copy Snapshot**.
- Select the destination region and initiate the copy.
- After the snapshot is copied, create a volume from it in the new region using the steps outlined above.

By following these steps, you can efficiently manage and scale your storage using EBS, ensuring your infrastructure remains flexible and robust.